NETWORK LAB REPORT

DHCP Server & DHCP Snooping Configuration

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1. Lab Topology & Objective

Objective: Configure a Cisco router as a DHCP server and implement DHCP Snooping on downstream switches to protect against rogue DHCP servers, ensuring end devices receive legitimate IP configurations.

Topology Layout:

- **R1 (Router):** Configured as the DHCP server. Interface G0/0 (192.168.1.1) connects to SW1's G0/2.
- **SW1 (Switch):** Core/Distribution layer switch connecting to R1 via G0/2 and downstream to SW2 via G0/1.
- **SW2 (Switch):** Access layer switch connecting to SW1 via G0/1. Connects to PC1 (F0/1), PC2 (F0/2), and PC3 (F0/3).
- **Network:** 192.168.1.0/24.

2. R1 Configuration (DHCP Server)

The router is configured to exclude the first 10 IP addresses and provide IP addresses from the `LAB-POOL` DHCP pool.

```
R1>en
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ip dhcp excluded-address 192.168.1.1 192.168.1.10
R1(config)#ip dhcp pool LAB-POOL
R1(dhcp-config)#network 192.168.1.0 255.255.255.0
R1(dhcp-config)#dns-server 8.8.8.8
R1(dhcp-config)#domain-name momoslab.com
R1(dhcp-config)#default-router 192.168.1.1
R1(dhcp-config)#lease 0 5 30
% Invalid input detected at '^' marker.

R1(dhcp-config)#lease?
% Unrecognized command
R1(dhcp-config)#
```

3. SW2 Configuration (Access Switch)

SW2 connects directly to the PCs. The access ports are assigned to VLAN 1, and DHCP snooping is enabled globally and for VLAN 1. Interface G0/1 (uplink to SW1) is configured as a trusted port.

```
SW2>en
SW2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
SW2(config)#int range f0/1 - 3
SW2(config-if-range)#switchport mode access
SW2(config-if-range)#switchport access vlan 1
SW2(config-if-range)#no shut
SW2(config-if-range)#ex
SW2(config)#ip dhcp snooping
SW2(config)#ip dhcp snooping vlan 1
SW2(config)#int g0/1
SW2(config-if)#ip dhcp snooping trust
SW2(config-if)#
```

4. SW1 Configuration (Distribution Switch)

SW1 is configured with DHCP snooping to secure DHCP traffic between the router and the access switch. G0/2 (uplink to R1) is configured as a trusted port.

```
SW1>en
SW1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
SW1(config)#ip dhcp snooping
SW1(config)#int g0/2
SW1(config-if)#ip dhcp snooping trust
SW1(config-if)#ex
SW1(config)#ip dhcp snooping vlan 1
SW1(config)#
```

DHCP Option 82 Issue & Resolution

Following the initial setup, PC1 attempted to request an IP address but failed. SW1 generated syslog messages indicating packets were being dropped due to Option 82/non-zero giaddr mismatches on untrusted ports.

```
SW1(config)#19:19:48: %DHCP_SNOOPING-5-DHCP_SNOOPING_NONZERO_GIADDR:
DHCP_SNOOPING drop message with non-zero giaddr or option82 value on
untrusted port, message type: DHCP DISCOVER, MAC sa: 0001.6432.B922
19:19:58: %DHCP_SNOOPING-5-DHCP_SNOOPING_NONZERO_GIADDR: DHCP_SNOOPING drop
message with non-zero giaddr or option82 value on untrusted port, message
type: DHCP DISCOVER, MAC sa: 0001.6432.B922
```

To resolve this standard Cisco packet tracer behavior with DHCP snooping, the Option 82 information insertion was disabled on SW1:

```
SW1(config)#no ip dhcp snooping information option
SW1(config)#
```

5. End Device Verification

After resolving the DHCP Snooping Option 82 issue on SW1, the PCs successfully negotiated DHCP leases from R1.

PC1 DHCP Request Log

```
C:\>ipconfig /renew
DHCP request failed.
C:\>DHCP request failed.
...
C:\>ipconfig /renew

   IP Address......................: 192.168.1.11
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: 192.168.1.1
   DNS Server......................: 8.8.8.8
```

PC2 DHCP Request Log

```
C:\>ipconfig /renew
Invalid Command.

C:\>ipconfig /renew

   IP Address......................: 192.168.1.12
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: 192.168.1.1
   DNS Server......................: 8.8.8.8
```

PC3 DHCP Request Log

```
C:\>ipconfig /renew

   IP Address......................: 192.168.1.13
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: 192.168.1.1
   DNS Server......................: 8.8.8.8
```